



INDIAN INSTITUTE OF INFORMATION TECHNOLOGY SONEPAT
भारतीय सूचना प्रौद्योगिकी संस्थान सोनीपत
(An Autonomous Institute of National Importance under Act of Parliament)

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Mid Sem-I
Branch: IT

Subject Name: Data Structure
2nd Semester

Subject code: ITC 203
M.M.-15 marks

Roll no 12312021
Time- 60 mint.

Note: (1) Attempt all the questions.

(2) All questions carry equal marks (3*5=15)

Q-1	Differentiate the terms Abstract Data Type, Data Structure and Algorithm. What do you understand by worst case complexity? Explain clearly.	(3 marks)
Q-2	Let a two-dimensional array $arr[20][10]$ with four words per memory cell. Let the base address of array arr be 100 and the first element is $arr[0][0]$. Find the address of $arr[11][5]$ using row-major order and Column-major order?	(3 marks)
Q-3	Analyze the time complexity of various array (n length) operations such as a. Accessing an element, b. Searching for an element c. Finding minimum and maximum d. Sorting an array.	(3 marks)
Q-4	What are the various Operations performed on the Stack? Write an algorithm for matching different parenthesis in an algebraic expression.	(3 marks)
Q-5	Convert the given postfix expression $12 + 3 * 4 5 * -$ into prefix and infix using stack. Discuss the time complexity of expression (length n) conversion and evaluation using stack.	(3 marks)

$12 + 3 * 4 5 * -$

End



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Mid Sem-I

Subject: OOP Using C++

Roll no. 12312021

Branch: IT

M.M.-15 marks

Time- 60 mint.

Note: All questions are compulsory.

Q-1	Discuss the role of constructors and destructors in C++ classes. Explain the differences between default constructors, parameterized constructors, copy constructors, and destructors. Provide code examples to illustrate the use of each type of constructor and destructor.	(3 marks)
Q-2	Describe the significance of access specifiers (public, private, and protected) in C++ classes. Explain how these access specifiers control the visibility of class members and discuss their impact on encapsulation and data hiding in object-oriented programming.	(3 marks)
Q-3	Explain the concept of the this pointer in C++ and its role in object-oriented programming. Describe how the this pointer is used to refer to the current object within member functions, and provide examples to illustrate its usage. Discuss the importance of the this pointer in resolving naming conflicts and accessing class members.	(3 marks)
Q-4	Explain the concept of dynamic memory allocation in C++ using the new and delete operators. Discuss the advantages and disadvantages of dynamic memory allocation compared to static memory allocation, and provide examples to illustrate their usage. Discuss memory management issues that may arise when using dynamic memory allocation.	(3 marks)
Q-5	Implement a C++ program to sort an array of integers using the bubble sort algorithm. Ensure that your program accepts user input for the array elements and displays the sorted array.	(3 marks)



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Mid Sem-I

Subject: **COMMUNICATION SKILLS (HUL 204)**

Branch: IT

M.-15 marks

Time- 60 min.

Semester II 2023-20027

Batch Roll no. ...12312021.....

Attempt All. All questions carry equal marks. (3 X 5 = 15)

1. What do you understand by the term 'communication'? Discuss various types of communication. With the help of suitable examples, discuss their relative advantages and disadvantages.
2. Write a detailed note on the various skills of communication. Discuss their strengths and weaknesses with respect to different areas of communication.
3. Elaborate on different Barriers to communication. Discuss the various types of barriers to communication and elaborate with the help of suitable examples how they pose issues and challenges to comprehension?



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Mid Sem-I
Branch: IT

Subject: Web Designing (ITC 205)
M.M.-15 marks

Roll no. 12312021..
Time- 60 mint.

Note: All questions are compulsory.

Q-1: (a)	An image named iiits.png is 200 pixels wide and 150 pixels high. Write an HTML5 statement using the width and height attributes of the img element to perform each of the following transformations: a) Increase the size of the image by 100 percent. b) Increase the size of the image by 50 percent. c) Change the width-to-height ratio to 2:1, keeping the width obtained in part (a).	(2marks)
(b)	Create a link to each of the following: a) The file index.html, located in the files directory. b) The file index.html, located in the text subdirectory of the files directory. c) The file index.html, located in the other directory in your parent directory. d) The President's email address (president@whitehouse.gov). e) The file named README in the pub directory of ftp.cdrom.com	(2marks)
Q-2	Write an HTML5 element (or elements) to accomplish each of the following tasks: a) Use a datalist to provide an autocomplete list for five states. b) Create a range input element that allows the user to select a number from 1 to 100. c) Specify that autocomplete should not be allowed for a form. Show only the form's opening tag. d) Use a mark element to highlight the second sentence in the following paragraph. Students were asked to rate the food in the cafeteria on a scale of 1 to 10. The average result was 7.	(4 marks)
Q-3	Make a layout template that contains a header and two paragraphs. Use float to line up the two paragraphs as columns side by side. Give the header and two paragraphs a border and/or a background color so you can see where they are.	(4 marks)
Q-4	Create three div elements of varying colors, each with a width and height of 200px. On the first box, add a dim grey box-shadow with a horizontal offset of 15px, a vertical offset of 15px and a blur radius of 20px. On the second box, add a dim grey box-shadow with a horizontal offset of -15px, a vertical offset of -15px and a blur radius of 30px. On the third box, add a dim grey box-shadow with a horizontal offset of 15px, a vertical offset of 15px and a blur radius of 10px.	(3 marks)

Roll No:

1 2 3 1 2 0 2 1



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B.Tech II Sem I Mid Semester Examination, 26th Feb, 2024

Mid Sem-I

Subject: Mathematics-II

Course Code: MAL-201

Branch: IT

Maximum Marks. -15 marks

Time- 60 min.

Note: All questions are compulsory.

Q-1	By changing the order of integration, evaluate $\int_0^a \int_{\frac{y^2}{a}}^y \frac{y}{(a-x)\sqrt{ax-y^2}} dx dy$.	(3marks)
Q-2	By change of variable, find the volume of the sphere $x^2 + y^2 + z^2 = a^2$.	(3marks)
Q-3	Find the directional derivative of $\nabla(\nabla f)$ at the point $(1, -2, 1)$ in the direction of the normal to the surface $xy^2z = 3x + z^2$, where $f = 2x^3y^2z^4$.	(3marks)
Q-4	Using Gauss divergent theorem, Evaluate $\iint_S \vec{F} \cdot \hat{n} ds$, where $\vec{F} = 4x\hat{i} - 2y^2\hat{j} + z^2\hat{k}$ and S is the surface bounding the region $x^2 + y^2 = 4$, $z = 0$ & $z = 3$.	(3marks)
Q-5	Using Green's Theorem, Evaluate the line integral $\int_C [(y - \sin x)dx + \cos x dy]$, where C is the plane triangle enclosed by the lines $y = 0$, $x = \frac{\pi}{2}$ and $y = \frac{2x}{\pi}$.	(3marks)



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Mid Sem-I

Subject: Digital Electronics (ECL-202)

Roll no ... 12312021

Branch: IT

M.M.-15 marks

Time- 60 mint.

Note: All questions are compulsory.

Q-1: (a)	Convert the decimal number 321 to binary and base-5 number system.	(2marks)															
(b)	Simplify the below function using Boolean's law $f(A, B, C) = ABC + AB'C' + AB'C + ABC' + A'B'C$	(2marks)															
Q-2	Compute the outputs w, x, y, and z for the inputs shown in the below table. <table><tr><th>S. No.</th><th>Input</th><th>Output</th></tr><tr><td>1</td><td>(0111)_{BCD}</td><td>(w)_{XS-3}</td></tr><tr><td>2</td><td>(0101)_{XS-3}</td><td>(x)_{BCD}</td></tr><tr><td>3</td><td>(1110)_{Binary}</td><td>(y)_{Gray}</td></tr><tr><td>4</td><td>(0110)_{Gray}</td><td>(z)_{Binary}</td></tr></table>	S. No.	Input	Output	1	(0111) _{BCD}	(w) _{XS-3}	2	(0101) _{XS-3}	(x) _{BCD}	3	(1110) _{Binary}	(y) _{Gray}	4	(0110) _{Gray}	(z) _{Binary}	(2 marks)
S. No.	Input	Output															
1	(0111) _{BCD}	(w) _{XS-3}															
2	(0101) _{XS-3}	(x) _{BCD}															
3	(1110) _{Binary}	(y) _{Gray}															
4	(0110) _{Gray}	(z) _{Binary}															
Q-3	Compare and explain radix complement and diminished radix complement in the decimal number system with examples.	(3 marks)															
Q-4	Simplify the expression $138 + 75$ using BCD Addition.	(3 marks)															
Q-5	Minimize the following function using K-Map and draw a circuit diagram for the minimized expression. $F = A'B'C' + AC' + A'B$	(3 marks)															